

TEZPUR UNIVERSITY

End-Semester Examination, Autumn 2016

CO 504 NATURAL LANGUAGE PROCESSING

Marks : 60

Time : 2 hours

The figures in the right-hand margin indicate marks for the individual question.

Answers should be concise and entire answer to a question should be together. Give examples and diagrams wherever appropriate. State assumptions wherever made.

Submit question no. 11 within first ten minutes.

1. A language can be seen as a scheme of using a symbols from a defined set in a controlled way to form sequences. Explain. 5
2. What are some common morphological phenomena in English? Why may we say that most Indian languages have a richer morphology than English? 3 + 2
3. Suppose $n_1, \dots, n_8$ are eight words in a language and the following are the unigram counts of these words in a corpus- $\{n_1:4, n_2:5, n_3:0, n_4:1, n_5:4, n_6:0, n_7:3, n_8:2\}$. Compute the unigram probabilities, and the add-one smoothed probabilities. 5
4. Consider the following sequence of words-
The *boys plays* cricket.
How can *unification of features* help in detecting the syntactic defect in the above sequence of words? 5
5. (a) What are some shortcomings of Probabilistic Context Free Grammar for syntax analysis? 3

(b) Give an example of analysis of a sentence using *dependency parsing*. 2

6. (a) Why is it not a good idea to have a First Order Predicate Calculus (FOPC) formula $TU_Student(x)$ to indicate that x is a TU_Student? What is a better way in FOPC? 2 + 1

(b) Express the following statement using FOPC- 3
Kamal ate fruits for breakfast at home.

7. (a) Identify the thematic roles of the arguments of the verb in the following sentence- 4

The wind blows dust from the road into our rooms.

(b) How do *metaphor* and *metonymy* pose a challenge to analysis of a sentence? 3

8. (a) The other words in a sentence can help in determining the sense of an ambiguous word in the sentence. Briefly describe how this can be carried out. 4

(b) How can the concept of *inverse document frequency* be used to identify *stop words* in a language? 2

9. How would you decide which referents do the pronouns refer to in the following sentences? 4

The organisers gave the actress an appreciation letter. They liked her performance in the play.

10. (a) How can the *deep semantic analysis* requirement of *inter-lingua based* translation be a limitation of the approach? 3

(b) Why is *approximate* machine translation usable in cross-language information retrieval? 2

11. Fill in the blanks 1 × 7

(a) Obtaining the word *government* from the word *govern* is an instance of _____ (derivation / inflection).

(b) *Car:vehicle* is an example of _____ (hyponymy / synonymy).

(c) *Grounding* is a phenomenon observed in _____ (monologue / dialogue).

(d) The sentences “Anil went to Guwahati. Vimal does not like cricket.” lack coherence. _____ (True / False)

(e) When translation is required between one another among several languages _____ (direct / interlingua based) translation is a more appropriate approach.

(f) Collocation feature vectors can help in _____ (word sense disambiguation/reference resolution).

(g) *Before he took the test Anil revised the class notes.* Here *he* is an/a _____ (anaphora/cataphora).